SOFTWARE ENGINEERING

Software Engineering Project: Building Real-World Software Systems

This semester project challenges teams of 3–5 students to develop a realistic and feasible software product while mastering professional software engineering processes. The emphasis is on **process, planning, and management** rather than coding complexity alone. A simple system managed professionally will score higher than a complex unmanaged one.

Project Scope & Focus

Teams will build a complete software product—whether a web app, mobile app, platform, system, or service—using industry-standard Agile methodologies. The project spans two major phases: initial planning and setup (Phase I), followed by incremental development through two working sprints (Phase II). Students must demonstrate proficiency in version control, Agile project management, system modeling, risk management, and testing practices.

Mandatory Professional Tools

- **Git + GitHub** for version control and team collaboration
- **Jira (free plan) or Trello** for Scrum/Agile management
- **Planning Poker tool** for story estimation
- **Diagram tool** (Draw.io, Figma, or Lucidchart free versions)

All students must provide repository links, board links, and clear evidence of consistent tool usage throughout the project lifecycle.

Course Instructors

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Evaluation Breakdown

Criteria	Weight
Quality of increments & functional delivery	25%
Agile process & Scrum tool usage	20%
Git usage & collaboration discipline	15%
Architecture & system modeling	15%
Final demo & presentation	15%
Testing & risk management	10%

PHASE I

Project Definition & Agile Setup

Deadline: End of March

Phase I establishes the foundation for successful project execution. Teams must define their project vision clearly, set up the complete Agile environment, and prepare for iterative development. This phase emphasizes thorough planning, stakeholder analysis, and the creation of a comprehensive product backlog that will guide all subsequent development work.

01

Project Vision & Scope Definition

Articulate the system need, objectives, and boundaries. Identify all stakeholders and their interests. Select your development approach (Scrum or Kanban) and define clear team roles.

02

Product Backlog Creation

Develop a minimum of 15–20 user stories with complete acceptance criteria for each. Use Planning Poker for estimation and prioritization. Prepare Sprint 1 planning with selected stories.

03

Requirements Documentation

Create a Software Requirements Document following simplified IEEE structure: introduction with system need and objectives, user requirements (functional and non-functional), high-level system architecture, and system models where applicable.

04

Tool Setup & Configuration

Establish GitHub repository with branching strategy, pull request workflow, and meaningful commits from all members. Configure Jira/Trello board with epics, user stories, priorities, and Sprint 1 definition.

Phase I Deliverables

Documentation

Vision & Requirements document (PDF format)

Repository Access

GitHub repository link with evidence of setup

Sprint Planning

Complete Sprint 1 plan with selected stories

Presentation

Short presentation of project vision and setup (10 minutes)

Project Board

Scrum board link (Jira/Trello)

Cost Estimation

Effort and cost estimation report

Success Criteria: Professional structure required for all documentation. Teams must demonstrate clear evidence of collaborative tool usage, with all members actively contributing to the repository and project board. A simple system managed professionally scores higher than a complex unmanaged one.

PHASE II

Incremental Development & Engineering Practice

Deadline: Week of May 11

Phase II focuses on implementing the system through **two working increments** using professional Agile practices. Each increment must deliver demonstrable functionality, with proper branching, pull requests, and testing. Teams will produce comprehensive engineering artifacts including system models, risk management plans, and testing documentation, culminating in a live demonstration of the fully working system.

Increment 1

Sprint planning, core features, branching workflow, demonstrable version

Increment 2

Additional features, refactoring, testing, complete system

Each increment builds upon the previous work, progressively delivering more complete functionality while maintaining code quality and system integrity through continuous integration practices.

Required Engineering Artifacts

System Modeling & Design

- **Use Case Diagram** showing system interactions
- **2 Detailed Use Cases** with complete specifications
- **2 Activity Diagrams** illustrating workflows
- **1 State Diagram** for key system states
- **2 Sequence Diagrams** showing object interactions
- **Architecture overview** with component relationships
- **Database design** (if applicable to your system)

Risk Management (RMMM Plan)

Identify and document at least 5–6 project risks with:

- Probability assessment
- Impact analysis
- Mitigation strategy for each risk

System Testing

Develop minimum 5–8 acceptance test cases including:

- Clear expected results
- Pass/fail criteria
- Test execution documentation

Final Submission Requirements

Working System

Fully functional system with **mandatory live demo**. System must demonstrate core features and handle real-world scenarios.

Updated Backlog

Complete sprint history visible in Jira/Trello with all completed stories, velocity tracking, and burndown charts.

Technical Documentation

All diagrams, risk documentation, testing documentation, and final structured report in professional format.

Team Presentation

20–25 minute presentation with **all members participating**. Demonstrate system, explain architecture, and discuss engineering decisions.

> *"The quality of your increments and the professionalism of your engineering process will determine your success. Focus on delivering working software through disciplined Agile practices, comprehensive documentation, and effective team collaboration."*

Software Engineering @Lebanese University

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